



CHAPTER

6

MEng 2132

Engineering Thermodynamics I

Lecture Notes:

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Entropy



Objective of the chapter

- ❖ Apply the second law of thermodynamics to processes.
- ❖ Define a new property called entropy to quantify the second-law effects.
- ❖ Establish the increase of entropy principle.
- ❖ Calculate the entropy changes that take place during processes for pure substances, incompressible substances, and ideal gases.

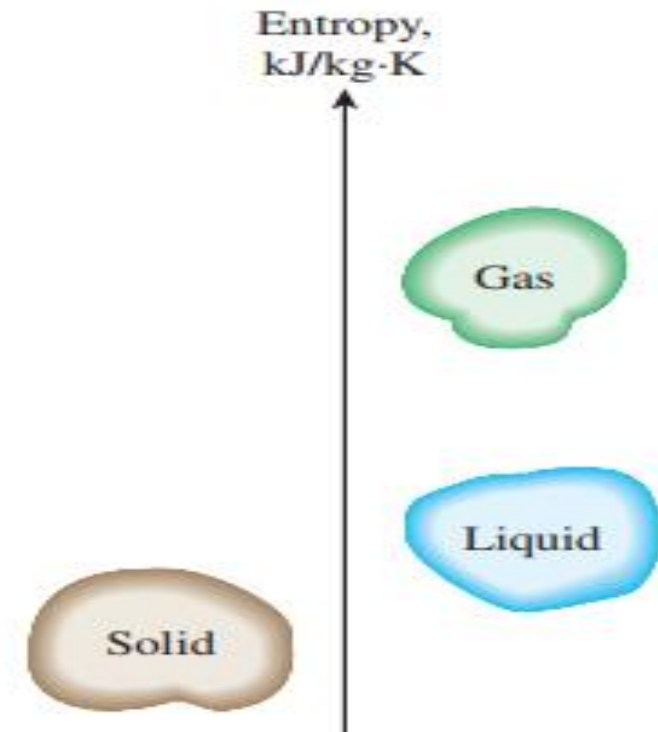


Objective of the chapter

- ❖ Examine a special class of idealized processes, called isentropic processes, and develop the property relations for these processes.
- ❖ Derive the reversible steady-flow work relations.
- ❖ Develop the isentropic efficiencies for various steady-flow devices.
- ❖ Introduce and apply the entropy balance to various systems

Entropy

- ❑ The Entropy can be viewed as a measure of molecular disorder, or molecular randomness.
- ❑ As a system becomes more disordered, the positions of the molecules become less predictable and the entropy increases.
- ❑ Thus, it is not surprising that the entropy of a substance is lowest in the solid phase and highest in the gas phase



The level of molecular disorder (entropy) of a substance increases as it melts or evaporates.

Statistical Mechanics Perspective: Entropy can be defined in terms of the number of microscopic configurations that correspond to a thermodynamic system's macroscopic state:

$$S = k \ln p$$

P =thermodynamic probability

K = the Boltzmann constant

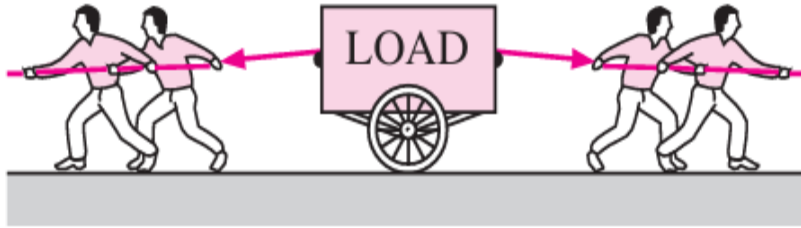


Entropy

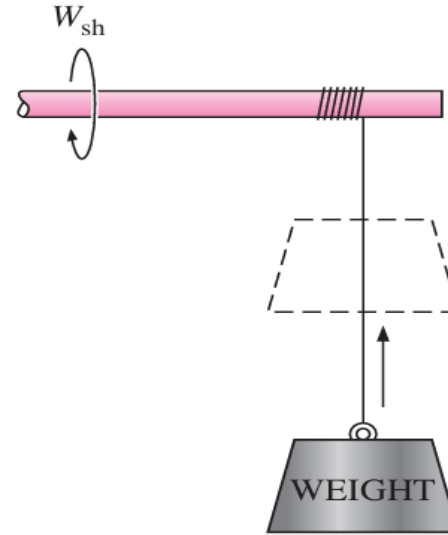
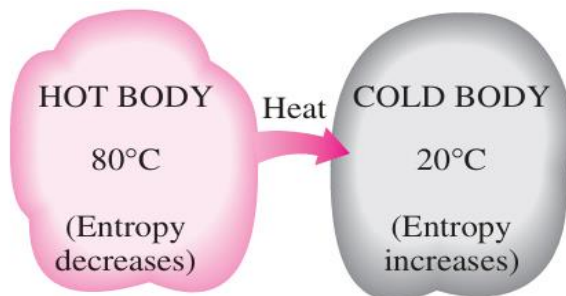
- ❑ The 2nd law states that process occur in a certain direction, not in any direction.
- ❑ It often leads to the definition of a new property called *entropy*, which is a quantitative measure of disorder for a system.
- ❑ Entropy can also be explained as a measure of the unavailability of heat to perform work in a cycle.
- ❑ This relates to the 2nd law since the 2nd law predicts that not all heat provided to a cycle can be transformed into an equal amount of work, some heat rejection must take place.

Statistical Mechanics Perspective: Entropy can be defined in terms of the number of microscopic configurations that correspond to a thermodynamic system's macroscopic state:

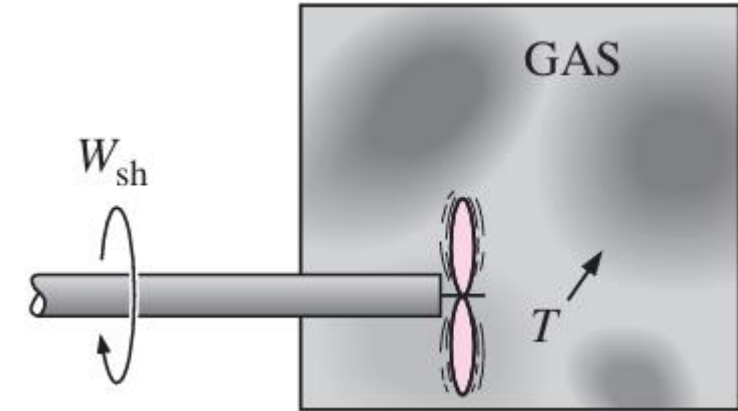
Entropy



Disorganized energy does not create much useful effect, no matter how large it is.



In the absence of friction, raising a weight by a rotating shaft does not create any disorder (entropy), and thus energy is not degraded during this process.



The paddle-wheel work done on a gas increases the level of disorder (entropy) of the gas, and thus energy is degraded during this process.

Clausius inequality

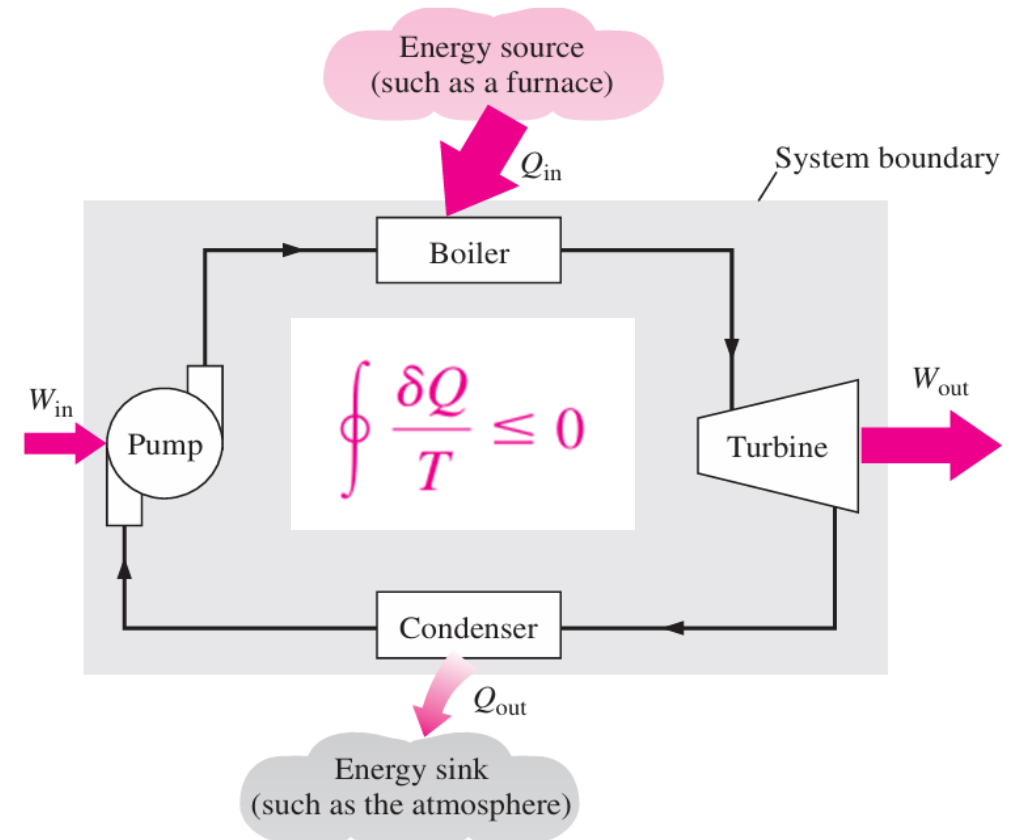
- Another important inequality that has major consequences in thermodynamics is the **Clausius inequality**. It was first stated by the German physicist R. J. E. Clausius (1822–1888), one of the founders of thermodynamics, and is expressed as



Rudolf Clausius (1822-1888)

“The energy of the universe is constant.”

“The entropy of the universe tends to a maximum.”



The Inequality of Clausius

$$\oint \frac{\delta Q}{T} \leq 0$$

- The inequality of Clausius is a consequence of the second law of thermodynamics. For reversible cycle
- **Q** is the heat transfer to or from the system.
- **T** is the absolute temperature at the boundary.
- The symbol $\oint$ is the cyclic integral

$$\oint \left(\frac{\delta Q}{T} \right)_{\text{int rev}} = 0$$

For reversible cycle

$$\oint \left(\frac{\delta Q}{T} \right) < = 0$$

For irreversible cycle

Entropy Change

- ❑ Clausius realized in 1865 that he had discovered a new thermodynamic property, and he chose to name this property **entropy**. It is designated S

The entropy change during a reversible process is defined as

$$dS = \frac{\delta Q_{net}}{T} \Big|_{\text{int rev}}$$

(kJ/K)

$$S_2 - S_1 = \int_1^2 \frac{\delta Q_{net}}{T} \Big|_{\text{int rev}}$$

- ❑ For a reversible, adiabatic process

$$dS = 0$$

$$S_2 = S_1$$

- ❑ The *reversible, adiabatic process* is called an **isentropic process**.

THE INCREASE OF ENTROPY PRINCIPLE

- Consider a cycle that is made up of two processes: **process 1-2**, which is arbitrary (reversible or irreversible), and **process 2-1**, which is internally reversible,

From the Clausius inequality,

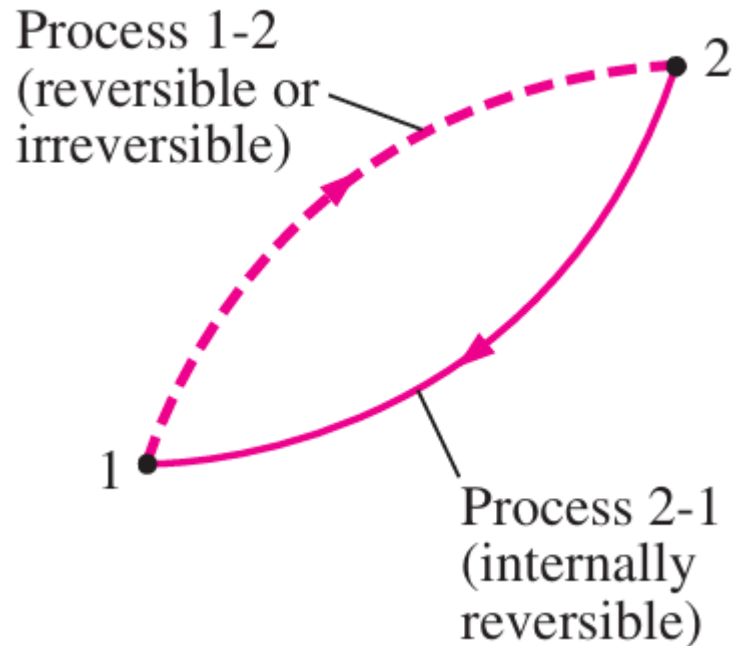
$$\oint \frac{\delta Q}{T} \leq 0$$

$$\int_1^2 \frac{\delta Q}{T} + \int_2^1 \left(\frac{\delta Q}{T} \right)_{\text{int rev}} \leq 0$$

$$\int_1^2 \frac{\delta Q}{T} + S_1 - S_2 \leq 0$$

which can be rearranged as
$$S_2 - S_1 \geq \int_1^2 \frac{\delta Q}{T}$$

$$\Delta S_{\text{sys}} = S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} + S_{\text{gen}}$$



THE INCREASE OF ENTROPY PRINCIPLE

$$\Delta S_{\text{sys}} = S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} + S_{\text{gen}}$$



Entropy
change



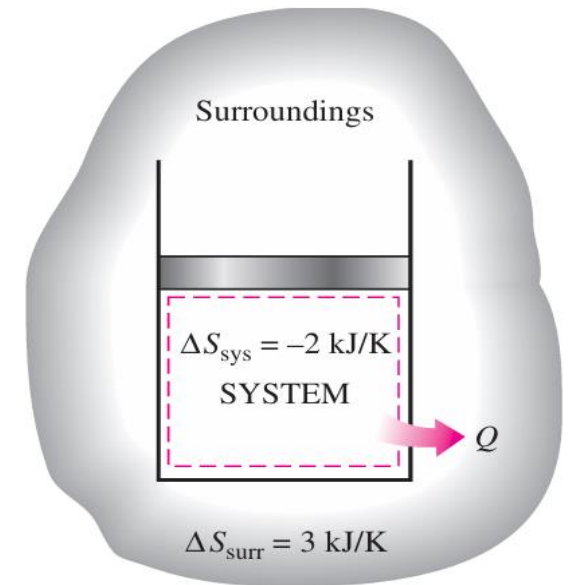
Entropy
transfer by
heat



Entropy generation

This is known as the increase of entropy principle.

$$S_{\text{gen}} \begin{cases} > 0 & \text{Irreversible process} \\ = 0 & \text{Reversible process} \\ < 0 & \text{Impossible process} \end{cases}$$



$$S_{\text{gen}} = \Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 1 \text{ kJ/K}$$

Some Remarks about Entropy

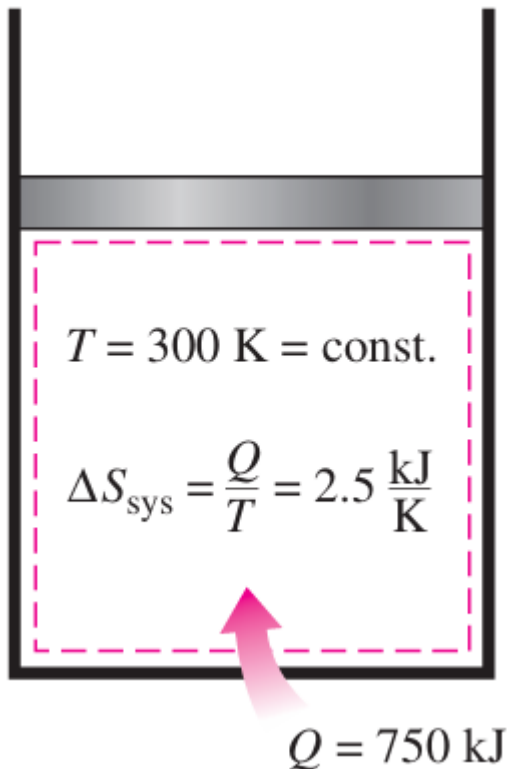
1. Processes can occur in a **certain direction only**, not in any direction. A process must proceed in the direction that complies with the increase of entropy principle, that is, **that is, $S_{\text{gen}} \geq 0$** .
2. Entropy is a **nonconserved** property, and there is **no such thing** as the **conservation of entropy principle**. Entropy is conserved during the idealized reversible processes only and **increases during all actual processes**.
3. The performance of engineering systems is degraded by the presence of irreversibilities, and entropy generation is a measure of the magnitudes of the irreversibilities present during that process. **The greater the extent** of irreversibilities, the **greater the entropy generation**. Therefore, entropy generation can be used as a quantitative measure of irreversibilities associated with a process.

Summary of Energy and Entropy

Aspect	Energy	Entropy
Definition	Capacity to do work	Measure of disorder
Conservation	Conserved (First Law)	Increases (Second Law)
Units	Joules (J)	Joules per Kelvin (J/K)
Directionality	Can be transferred/converted	Indicates irreversibility
Nature	Associated with work	Associated with disorder

Example 6.1

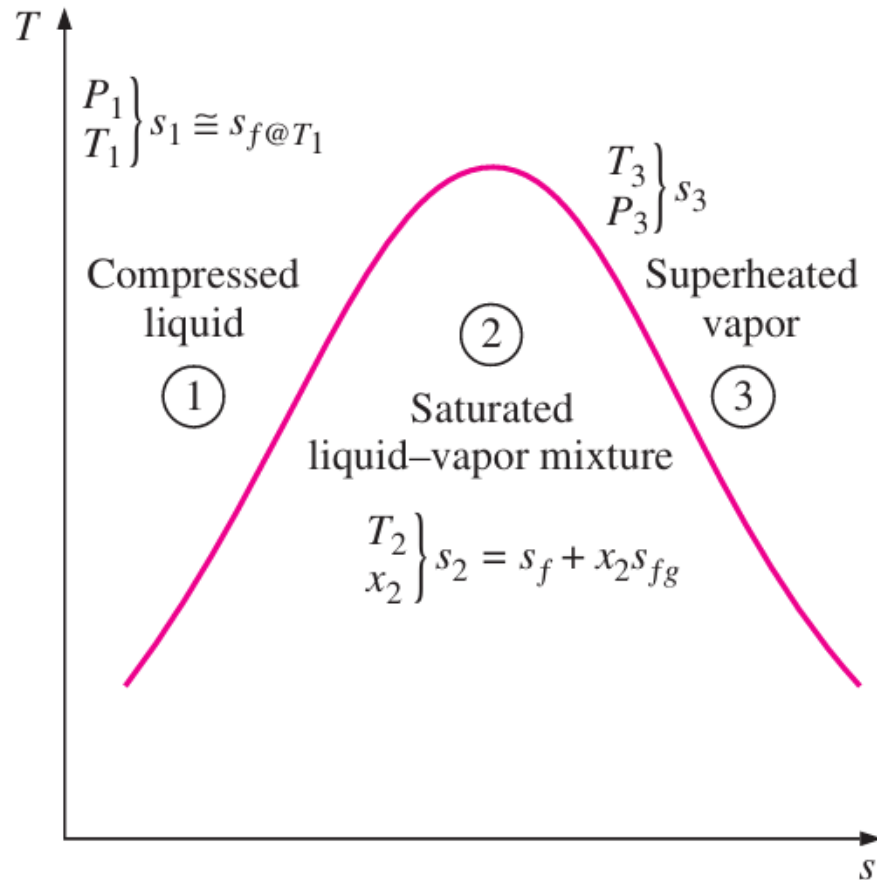
A piston–cylinder device contains a liquid–vapor mixture of water at 300 K. During a constant-pressure process, 750 kJ of heat is transferred to the water. As a result, part of the liquid in the cylinder vaporizes. Determine the entropy change of the water during this process.



Solution

$$\Delta S_{\text{sys, isothermal}} = \frac{Q}{T_{\text{sys}}} = \frac{750 \text{ kJ}}{300 \text{ K}} = \mathbf{2.5 \text{ kJ/K}}$$

ENTROPY CHANGE OF PURE SUBSTANCES



$$s = s_f + x s_{fg} \quad (\text{kJ/kg} \cdot \text{K})$$

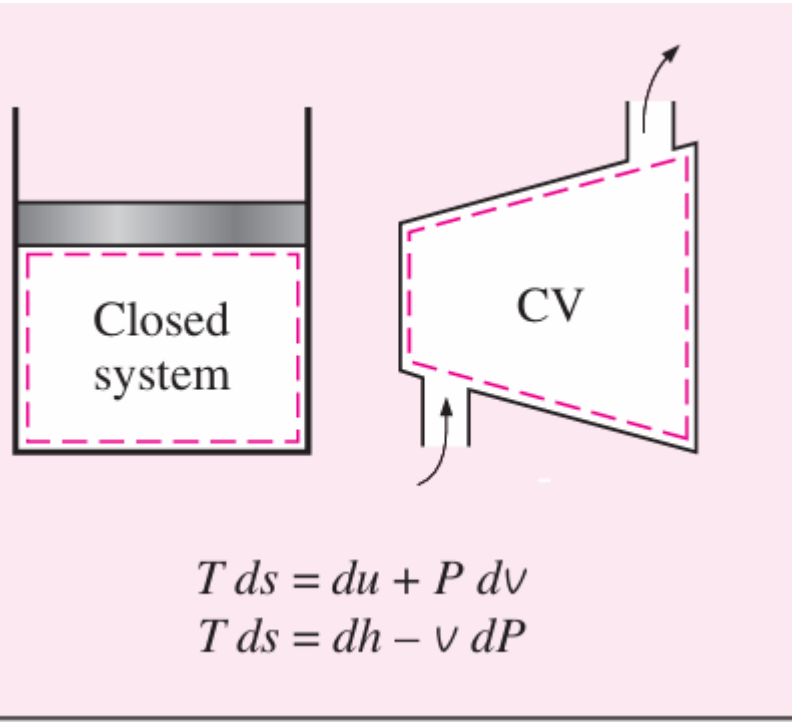
In the absence of compressed liquid data, the entropy of the compressed liquid can be approximated by the entropy of the saturated liquid at the given temperature:

$$s@T,P \cong s_f@T \quad (\text{kJ/kg} \cdot \text{K})$$

The entropy change of a specified mass m (a closed system) during a process is simply

$$\Delta S = m\Delta s = m(s_2 - s_1) \quad (\text{kJ/K})$$

THE Tds RELATIONS



$$\delta Q_{\text{int rev}} - \delta W_{\text{int rev,out}} = dU \longrightarrow$$

Energy balance for closed system

$$\delta Q_{\text{int rev}} = T dS$$

$$\delta W_{\text{int rev,out}} = P dV$$

$$T dS = dU + P dV \quad (\text{kJ})$$

$$T ds = du + P dv \quad (\text{kJ/kg}) \longrightarrow$$

First Tds equation

The second Tds equation is obtained by eliminating du from above by using the definition of enthalpy ($h=u+ Pv$):

$$h = u + Pv$$

From first Tds equ.

$$\left. \begin{array}{l} \longrightarrow dh = du + P dv + v dP \\ \longrightarrow T ds = du + P dv \end{array} \right\} T ds = dh - v dP \longrightarrow$$

Second Tds equation



Entropy Change and Isentropic Processes

The entropy-change and isentropic relations for a process can be summarized as follows:

i. Pure substances:

Any process: $\Delta s = s_2 - s_1$ (kJ/kg·K)

Isentropic process: $s_2 = s_1$

ii. Incompressible substances (liquids and solids):

Any process: *Liquids, solids:* $s_2 - s_1 = \int_1^2 c(T) \frac{dT}{T} \cong c_{\text{avg}} \ln \frac{T_2}{T_1}$ (kJ/kg·K)

Isentropic process: $T_2 = T_1$

iii. Ideal gases:

a) constant specific heats (approximate treatment):

for all process

$$s_2 - s_1 = C_{v,av} \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}$$

$$s_2 - s_1 = C_{p,av} \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$$

$$TV^{k-1} = \text{constant}$$

$$TP^{(1-k)/k} = \text{constant}$$

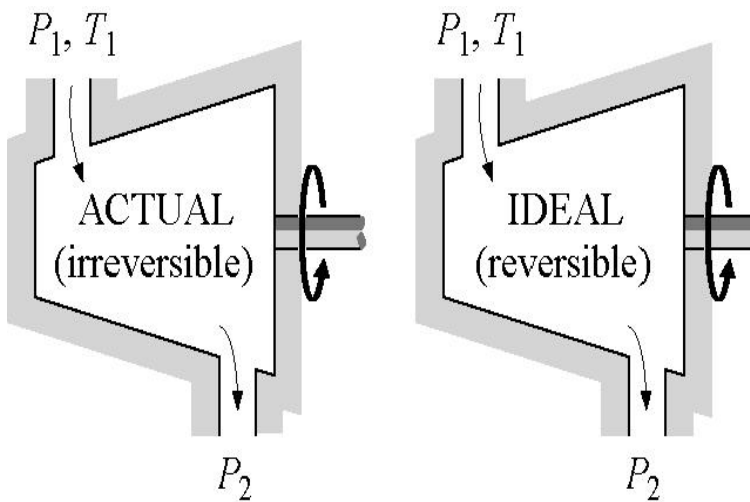
$$PV^k = \text{constant}$$

for *isentropic process*

$$\left(\frac{T_2}{T_1} \right)_{s=\text{const.}} = \left(\frac{v_1}{v_2} \right)^{k-1} \quad \left(\frac{T_2}{T_1} \right)_{s=\text{const.}} = \left(\frac{P_2}{P_1} \right)^{(k-1)/k} \quad \left(\frac{P_2}{P_1} \right)_{s=\text{const.}} = \left(\frac{v_1}{v_2} \right)^k$$

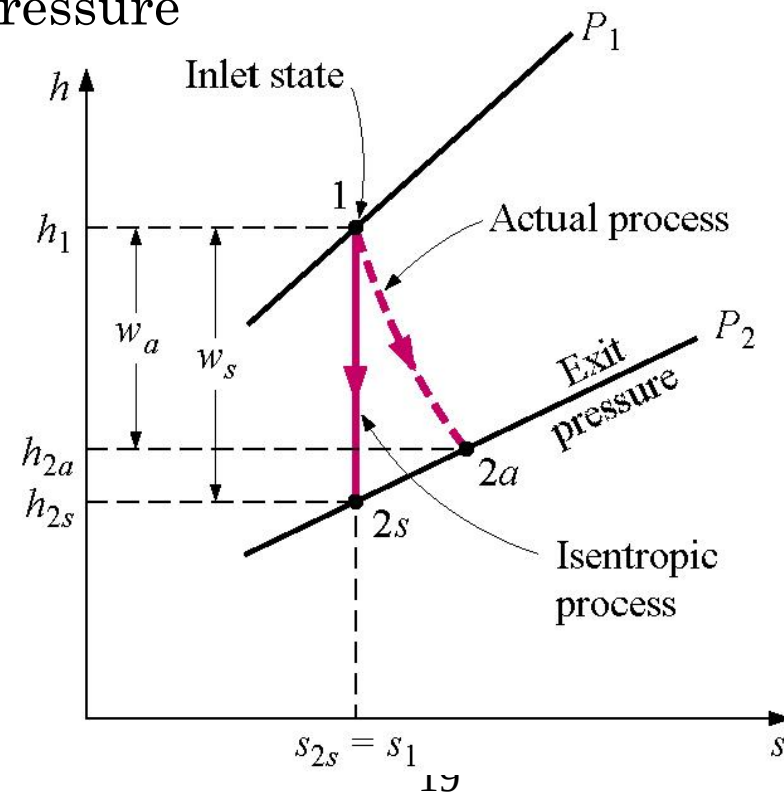
Isentropic Efficiency for Turbine

- The desired output of a turbine is the work produced, and the isentropic efficiency of a turbine is defined as the ratio of the actual work output of the turbine to the work output that would be achieved if the process between the inlet state and the exit pressure were isentropic:



$$\eta_T = \frac{\text{Actual turbine work}}{\text{Isentropic turbine work}} = \frac{w_a}{w_s}$$

$$\eta_T \equiv \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

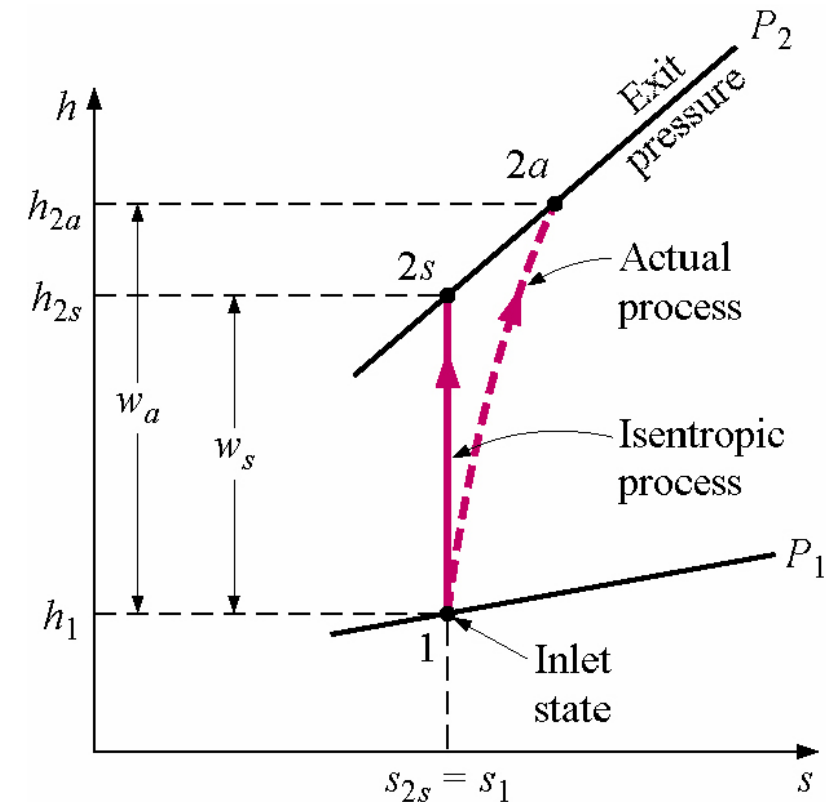
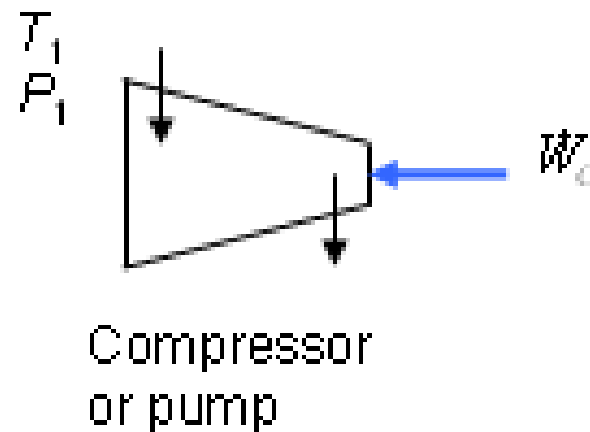


Isentropic Efficiency for Compressor

The isentropic efficiency of a compressor is defined as the ratio of the work input required to raise the pressure of a gas to a specified value in an isentropic manner to the actual work input:

$$\eta_c = \frac{\text{Isentropic compressor work}}{\text{Actual compressor work}} = \frac{w_s}{w_a}$$

$$\eta_c \equiv \frac{h_{2s} - h_1}{h_{2a} - h_1}$$



Example 6.2

Steam at 1 MPa, 600°C, expands in a turbine to 0.01 MPa. If the process is isentropic, find the final temperature, the final enthalpy of the steam, and the turbine work.

Solution:

mass balance : $\dot{m}_1 = \dot{m}_2 = \dot{m}$

energy balance

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 + \dot{W}_{out}$$

$$\dot{W}_{out} = \dot{m}(h_1 - h_2)$$

State1

$$\left. \begin{array}{l} P_1 = 1 \text{ MPa} \\ T_1 = 600^\circ \text{C} \end{array} \right\} \begin{array}{l} \text{superheated} \\ h_1 = 3698.6 \frac{\text{kJ}}{\text{kg}} \\ s_1 = 8.0311 \frac{\text{kJ}}{\text{kg.K}} \end{array}$$

- ❖ Since that the process is isentropic, $s_2 = s_1$

State 2

$$\left. \begin{array}{l} P_2 = 0.01 \text{ MPa} \\ s_2 = 8.0311 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \end{array} \right\} \begin{array}{l} \text{sat. mixture} \\ x_2 = 0.984 \end{array}$$

$$\begin{aligned} h_2 &= 191.8 + 0.984(2392.1) \\ &= 2545.6 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

$$T_2 = T_{\text{sat}@P_2} = 45.81^\circ \text{C}$$

- ❖ Work of turbine

$$\begin{aligned} W_{\text{out}} &= h_1 - h_2 \\ &= 3698.6 - 2545.6 \\ &= \underline{\underline{1153 \frac{\text{kJ}}{\text{kg}}}} \end{aligned}$$



Example 6.3

Steam at 1 MPa, 600°C, expands in a turbine to 0.01 MPa. The isentropic work of the turbine is 1152.2 kJ/kg. If the isentropic efficiency of the turbine is 90 percent, calculate the actual work. Find the actual turbine exit temperature or quality of the steam.

Solution:

❖ Theoretically:

$$\eta_{isen,T} = \frac{w_a}{w_s} = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

$$\begin{aligned} w_a &= \eta_{isen,T} \times w_s \\ &= 0.9(1153) \\ &= 1037.7 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

State 1

$$\left. \begin{array}{l} P_1 = 1 \text{ MPa} \\ T_1 = 600^\circ \text{C} \end{array} \right\} \begin{array}{l} h_1 = 3698.6 \frac{\text{kJ}}{\text{kg}} \\ s_1 = 8.0311 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \end{array}$$

State 2s

$$\left. \begin{array}{l} P_2 = 0.01 \text{ MPa} \\ s_{2s} = s_1 = 8.0311 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \end{array} \right\} \begin{array}{l} \text{sat. mixture} \\ x_{2s} = 0.984 \\ h_{2s} = 2545.6 \frac{\text{kJ}}{\text{kg}} \end{array}$$

❖ Obtain h_{2a} from W_a

$$w_a = h_1 - h_{2a}$$

$$\begin{aligned} h_{2a} &= h_1 - w_a \\ &= 2660.9 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

State 2a

$$\left. \begin{array}{l} P_2 = 0.01 \text{ MPa} \\ h_{2a} = 2660.9 \frac{\text{kJ}}{\text{kg}} \end{array} \right\} \begin{array}{l} \text{superheated} \\ T_{2a} = \underline{\underline{86.85^\circ \text{C}}} \end{array}$$



Example 6.4

Air enters a compressor and is compressed adiabatically from 0.1 MPa, 27°C, to a final state of 0.5 MPa. Find the work done on the air for a compressor isentropic efficiency of 80 percent.

❖ For isentropic process of IGL

$$\left(\frac{T_{2s}}{T_1}\right) = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}$$

$$T_{2s} = (27 + 273) \left(\frac{0.5}{0.1}\right)^{0.4/1.4}$$

$$= 475.4 \text{ K}$$

❖ Then

$$W_{c,s} = 1.005(475.4 - 300)$$

$$= 176 \frac{\text{kJ}}{\text{kg}}$$

$$W_{c,a} = \frac{W_{c,s}}{\eta_{isen,c}} = \underline{\underline{220 \frac{\text{kJ}}{\text{kg}}}}$$

Solution:

❖ From energy balance

$$\dot{W}_{c,s} = \dot{m}(h_{2s} - h_1)$$

$$W_{c,s} = \frac{\dot{W}_{c,s}}{\dot{m}} = h_{2s} - h_1$$

$$= C_P (T_{2s} - T_1)$$



Entropy balance

- The property entropy is a measure of molecular disorder or randomness of a system, and the second law of thermodynamics states that entropy can be created but it cannot be destroyed.

$$\left(\begin{array}{c} \text{Total} \\ \text{entropy} \\ \text{entering} \end{array} \right) - \left(\begin{array}{c} \text{Total} \\ \text{entropy} \\ \text{leaving} \end{array} \right) + \left(\begin{array}{c} \text{Total} \\ \text{entropy} \\ \text{generated} \end{array} \right) = \left(\begin{array}{c} \text{Change in the} \\ \text{total entropy} \\ \text{of the system} \end{array} \right) \quad S_{\text{in}} - S_{\text{out}} + S_{\text{gen}} = \Delta S_{\text{system}}$$

- The entropy balance relation above can be stated as: the entropy change of a system during a process is equal to the net entropy transfer through the system boundary plus the entropy generated within the system.

Entropy balance

- Entropy balance for any system undergoing any process can be expressed more explicitly as.

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

(kJ/K) or, in the **rate form**, as

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{dS_{\text{system}}/dt}_{\text{Rate of entropy generation}}$$

Closed Systems

$$\sum \frac{Q_k}{T_k} + S_{\text{gen}} = \Delta S_{\text{system}} = S_2 - S_1 \quad (\text{kJ/K})$$

➤ Control Volumes

$$\sum \frac{Q_k}{T_k} + \sum m_i s_i - \sum m_e s_e + S_{\text{gen}} = (S_2 - S_1)_{\text{CV}} \quad (\text{kJ/K})$$

or, in the rate form, as

$$\sum \frac{\dot{Q}_k}{T_k} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{S}_{\text{gen}} = dS_{\text{CV}}/dt \quad (\text{kW/K})$$

Steady-flow:

$$\dot{S}_{\text{gen}} = \sum \dot{m}_e s_e - \sum \dot{m}_i s_i - \sum \frac{\dot{Q}_k}{T_k}$$



Supplementary Problems

1. The radiator of a steam heating system has a volume of 20 L and is filled with the superheated water vapor at 200 kPa and 150°C. At this moment both inlet and exit valves to the radiator are closed. After a while the temperature of the steam drops to 40°C as a result of heat transfer to the room air. Determine the entropy change of the steam during this process.

[-0.132 kJ/.K]

2. A heavily insulated piston-cylinder device contains 0.05 m³ of steam at 300 kPa and 150°C. Steam is now compressed in a reversible manner to a pressure of 1 MPa. Determine the work done on the steam during this process.

[16 kJ]

3. A piston –cylinder device contains 1.2 kg of nitrogen gas at 120 kPa and 27°C. The gas is now compressed slowly in a polytropic process during which $PV^{1.3}=\text{constant}$. The process ends when the volume is reduced by one-half. Determine the entropy change of nitrogen during this process.

[-0.0617 kJ/kg.K]



4. Steam enters an adiabatic turbine at 8 MPa and 500°C with a mass flow rate of 3 kg/s and leaves at 30 kPa. The isentropic efficiency of the turbine is 0.90. Neglecting the kinetic energy of the steam, determine (a) the temperature at the turbine exit and (b) the power output of the turbine.

[69.09°C, 3054 kW]

5. Refrigerant-R134a enters an adiabatic compressor as saturated vapor at 120 kPa at a rate of 0.3 m³/min and exits at 1 MPa pressure. If the isentropic efficiency of the compressor is 80 percent, determine (a) the temperature of the refrigerant at the exit of the compressor and (b) the power input, in kW. Also, show the process on a T-s diagram with respect to the saturation lines.

[58.9°C, 1.70 kW]



THANK YOU.. 😊